

Fake Currency Detection Using Machine Learning and Image Processing

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ABSTRACT

The paper addresses the issue of determining whether the given sample of currency is fake. Based on the colors, width, and serial numbers indicated, various conventional techniques and approaches are available for identifying fake currency. Various machine learning techniques are proposed by image processing in the advanced period of computer science and high computing approaches, which provides 99.9% accuracy for the fake identification of the currency. Methods for detection and recognition that go beyond algorithms include entities like color, shape, paper width, and note image filtering. This paper suggests utilising K-Nearest Neighbours, followed by image processing, to identify counterfeit money. The banknote authentication dataset used in this was developed using advanced computational and mathematical techniques, providing accurate data and information about the entities and properties related to the currency. Machine learning algorithms and image processing are used in data processing and data extraction to obtain the desired outcome and accuracy.

Keywords: Accuracy, data extraction, features extraction, image processing, K-Nearest

INTRODUCTION

The majority of people in this century are familiar with technology and how it operates, but many still engage in illegal activities. One of these operations is the creation of fake currency, which is used to trick people. This proposal focuses on this unlawful practice and attempts to present a remedy for it. According to a report, fake currency continues to be the primary source of counterfeit instances in India. A system that may be integrated into electronic devices is suggested in order to stop this fraudulent conduct. It will immediately identify the fake note when it is scanned by the device. KNN is one of

the strategies being investigated that has been used in the past and will be employed in the proposed system with improved accuracy. A new data point is classified using the K-nearest neighbours (KNN) method, which stores all the existing data and organises it based on similarity. This means that using the KNN algorithm, fresh data can be quickly and accurately categorised into appropriate categories [6]. The distance metric most frequently employed is the Euclidean distance. After that, it distributes the point among the class's k closest neighbours (where k is an integer).

The primary computation for k-NN is the online” searching” for the k nearest neighbours of a particular testing case since it does not require the offline training phase. 1-NN is typically used as a baseline for the other classifiers even though using different k values is likely to generate different classification results because it can provide reasonable classification performances in a variety of pattern classification challenges.

A fake currency detection system not only stops the spread of counterfeit money, but it also encourages merchants to accept and process payments in cash, prevents people from using fake currency, and guarantees that money is moving through the economy properly [3].

RELATED WORK

Different kinds of research and study have been done in the past at various times. Different improvements and advancements were seen[1]. In earlier investigations, fake note identification data was gathered using professional cameras, but the accuracy of those data was fair to good thanks to straightforward machine learning techniques. For detecting fraudulent notes, K nearest neighbour techniques has traditionally been utilised. The larger the data size, the slower the systems were. After that, a method was developed to classify the accuracy and rate of recognition using certain improvements to machine learning algorithms and deep learning principles[12].

Due to high and big data sets, data sets were becoming warped, and even if the precision was 98%, it was not very effective. Earlier, just open cv and python

were used to carry out all of these detections, but increasingly, data were collected using contemporary deep learning approaches with a count of 100 photos per denomination and then measured[11].The accuracy of the training and test sets was evaluated. As a result, compared to other ways, the efficiency of the chain type increases. The system made advantage of the transfer learning concept. Another issue that required considerably more development was the noise, which was also recorded. A Convolutional neural network was then introduced into the measuring process to help with error reduction. The general analysis of loss trends focused on training loss (TL) and validation loss (VL). Training accuracy (TA) was typically used to analyse accuracy trends. The algorithms of effective machine learning, Deep convolutional neural networks, and image processing are used in 2021 to detect fake notes. It has demonstrated that efficiency is at its highest level now.

METHODOLOGY

Using an industrial camera, high-quality photographs of both genuine and fake cash were gathered to create the data set[2]. Wavelet Transform was applied to these images to extract characteristics from the obtained images. Following the Wavelet Transformation, the attributes collected were Variance, Skewness, Kurtosis, entropy, and Class of the currency. First, four of the five derived attributes are continuous and describe the characteristics of the note, while the fifth property, ”Class,” assigns a value of 0 to counterfeit currency and a value of 1 to real currency[9].

Normalizing the Dataset

After investigating the data set, it was discovered that the data needed to be scaled in order to prevent the model from favouring one particular feature over another. In order to do this, it has been determined to limit each feature's range between 0 and 1. To accomplish this, the preprocessing library's MinMaxScaler is imported, and it applies the fit and transform functions to the data set's features.

Algorithms used to Train the Model

Data training can be done using a variety of functionalities and algorithms. It has been determined to compare the operation of 3 algorithms and conduct a comparative study on them for the purposes of this project.

- K-Nearest Neighbours
- Support Vector Classifier
- Gradient Boosting Classifier

K-nearest neighbours

It is a supervised algorithm for lazy learning. It is referred to as lazy because it uses all the data for training while classification and lacks a dedicated training phase. It is mostly employed for classification and prediction issues. The algorithm categorises a given data point by examining its nearest neighbour and giving each one a weight based on the distance between them. The closer data points are given more weight when the distance is Euclidian, Manhattan, etc. There are a few benefits, such as the ease of implementation, flexibility in terms of feature kinds, and ease of handling many classes. The disadvantage is that testing requires a lot of computational work, which makes the algorithm slow.

KNN is an excellent alternative for the data set used because it isn't very large, making computation time less of a concern. Since it is non-parametric, there is no need to be concerned about the distribution of the data, and the decision boundary can take any shape.

Support vector classifier

It is a supervised machine learning technique that is primarily used for regression and classification issues. Each point in this procedure is plotted in an n-dimensional plane with its value serving as the sample's data point. The classification is then carried out by identifying the hyper-plane that divides the classification into the best subsets. This approach has a number of benefits, including good performance for non-linear decision limits, economical memory utilisation, and versatility. The algorithm is likely to perform poorly if the nu.

As our data collection is divided into two groups and has less features than samples, this approach may be used to it. Additionally, as was said previously, even 2 characteristics allow us to distinguish between genuine and fake currency. We should therefore obtain better results after projecting to a higher dimensional plane.

Gradient boosting classifier

It is a group of ineffective learners who, when combined, contribute to produce an extremely accurate prediction model. Algorithms that produce models with accuracy that is only marginally superior to response selection at random called poor

learners. So, to create a predictive model, they are organised using a decision tree-like model that asks layers of yes- or-no questions. Its benefits include being simple to understand and apply, capable of accurately capturing non-linear function dependencies, incredibly versatile and customisable, and finally capable of building a strong model utilising the output of numerous weak learners. However, if the ensemble is huge, they can take a lot of time and require a lot of memory.

This approach can be applied to our data collection because layered classification questions may be posed and answered for the data. Additionally, because there are fewer features, training and testing won't take as much time.

Implementation

The steps that we'll take for this project are as follows:

- Data set preprocessing and normalisation.
- Applying the K-fold cross validation procedure, divide the data.
- Using the techniques discussed above, training the model.
- Determine the performance metrics.

The data set was loaded and preprocessed at first. Data normalisation was necessary, as stated in the sections above, and it was accomplished using the MinMaxScaler tool found in the Sklearn preprocessing toolkit.

Independent and Dependent Variables- As previously discussed, the data set consists of 5 columns, 4 of which are the features obtained after the Waveform transform,

while the fifth column is the class that separates fake from authentic notes. For this research, the class was the target or dependent variable, while the first four attributes variance, skewness, kurtosis, and entropy were used as independent or input variables. After normalisation and preprocessing, data is divided. To divide the data, we used the Kfold functionality offered by sklearn.modelselection. Utilising this library serves the objective of improving one's understanding of the algorithm. The method is executed with each block of the data set acting as the test set after the data has been divided into k blocks. Due to the fact that each sample is used for both training and testing, this aids in acquiring a thorough knowledge of the model. The fact that our data set was small and didn't require much extra time for training with this method was another reason we chose to use it. We then started creating the model after dividing the data into training and testing sets.

Building the Model

The necessary classifiers for this project were loaded from the Sklearn library. We developed a function that returned a trained model with its performance measure scores after taking parameters such as the training and testing data sets and procedure. This process was done for each fold, and the results were tallied for each pass in a list. Lastly, a confusion matrix for each method is created using the cross-validation predict module. Lastly, a graphic of the findings was created to show how the algorithms performed [4].

Performance Measures

Performance metrics are used to verify if the model is accurate. Accuracy,

precision, and f-score are performance measurements for this project. For each performance measure, graphs were created, and the outcomes were compared. For this project, the k-fold cross-validation technique is employed. The KFold and cross-validation predict module from the Sklearn model selection library was mentioned above. The data set is divided into the number of "k" blocks that the user specifies, and the algorithm is performed once for each block as a test set 10 for the value of k. Following model construction, it is evaluated using the performance metrics mentioned above.

CONCLUSION

We can conclude that all three algorithms were incredibly accurate at categorising notes as genuine or counterfeit based on the used data set after applying them and analysing the results obtained. However, as was mentioned in the section above, KNN fared better than the other two. Only 2 fake notes were classified wrongly, giving it a classification accuracy of 99.9%. The very modest size of the data set used, however, limits this result. To expand on this, we suggest creating a much larger data set including images of actual and counterfeit cash notes that are taken from the real world. This will assist in offering a model that is lot more realistic [10].

REFERENCES

1. Kapare, D. V., Lokhande, S., & Kale, S. (2013). Automatic Cash Deposite Machine With Currency Detection Using Fluorescent And UV Light. *Int. J. Comput. Eng. Res*, 3(309-311).
2. Rathore, M. N., & Sagar, J. A. Review on Fake currency detection using feature extraction. *vol, 10*, 407-411.
3. Santhanam, K., Sekaran, S., Vaikundam, S., & Kumarasamy, A. M. (2013, September). Counterfeit currency detection technique using image processing, polarization principle and holographic technique. In *2013 Fifth International Conference on Computational Intelligence, Modelling and Simulation* (pp. 231-235). IEEE.
4. Ponishjino, P., Antony, K., Kumar, S., & JebaKumar, S. (2017, April). Bogus currency authorization using HSV techniques. In *2017 International conference of Electronics, Communication and Aerospace Technology (ICECA)* (Vol. 1, pp. 179-183). IEEE.
5. Zhang, Q., & Yan, W. Q. (2018, November). Currency detection and recognition based on deep learning. In *2018 15th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)* (pp. 1-6). IEEE.
6. Upadhyaya, A., Shokeen, V., & Srivastava, G. (2018, December). Analysis of counterfeit currency detection techniques for classification model. In *2018 4th International Conference on Computing Communication and Automation (ICCCA)* (pp. 1-6). IEEE.
7. Arya, S., & Sasikumar, M. (2019, March). Fake currency detection. In *2019 International Conference on Recent Advances in Energy-efficient Computing and Communication (ICRAECC)* (pp. 1-4). IEEE.

8. Kumar, A., & Kumar, A. (2020, December). Dog breed classifier for facial recognition using convolutional neural networks. In *2020 3rd International Conference on Intelligent Sustainable Systems (ICISS)* (pp. 508-513). IEEE.
9. Alimushwan, M. M., Mohaimin, A., Islam, R., & Chowdhury, S. (2016). *Fake currency detection using image processing method* (Doctoral dissertation, BRAC University).
10. Shende, M. N., & Patil, P. P. (2018). A Review on Fake Currency Detection using Image Processing. *Int. J. Futur. Revolut. Comput. Sci. Commun. Eng.*, 4(1), 391-393.
11. Agasti, T., Burand, G., Wade, P., & Chitra, P. (2017, November). Fake currency detection using image processing. In *IOP Conference Series: Materials Science and Engineering* (Vol. 263, No. 5, p. 052047). IOP Publishing.
12. Yadav, B. P., Patil, C. S., Karhe, R. R., & Patil, P. H. (2014). An automatic recognition of fake Indian paper currency note using MATLAB. *Int. J. Eng. Sci. Innov. Technol*, 3, 560-566.